

Predictive Modeling for Course Demand and Revenue Forecasting on EduPro

Devansh Adik Wable

Abstract

Online learning platforms encounter significant operational and strategic hurdles due to fluctuating course enrollment demand and unpredictable revenue streams. This study presents a comprehensive machine learning framework designed to forecast future course enrollment demand and revenue for EduPro, a commercial e-learning platform. Utilizing a multi-relational dataset comprising 3,000 users, 60 teachers, 60 courses, and 10,000 transaction records aggregated into 60 course-level analytical observations, we developed and evaluated several predictive paradigms.

For enrollment demand forecasting, the Random Forest model achieved a Mean Absolute Error (MAE) of 6.130417 students and a Root Mean Squared Error (RMSE) of 7.322472 students, yielding the lowest RMSE and highest R^2 among evaluated enrollment models, while Gradient Boosting achieved the lowest MAE (6.050173 students). Due to low variance in course-level student enrollment across the platform, the enrollment models produced negative coefficients of determination (Random Forest $R^2 = -0.236758$). In contrast, the Gradient Boosting model proved highly effective for revenue forecasting, attaining an MAE of \$814.003007, an RMSE of \$1,457.664702, and an R^2 score of 0.960063. Feature importance analysis indicates that historical financial metrics—specifically Past Revenue (49.97%), Course Price (25.35%), and Past Revenue per Enrollment (20.44%)—account for 95.76% of the predictive weight in the revenue model.

Aggregate platform-level forecasts matched actual outcomes closely, achieving 99.371784% aggregate forecast accuracy for enrollment demand (predicted 2,464.385 vs. actual 2,449.00 students) and 98.022109% aggregate forecast accuracy for revenue (predicted \$222,744.35 vs. actual \$227,238.89). The analysis also highlighted a structural split: 38 free courses generated 64.15% of enrollments (1,571 students) but zero direct revenue, whereas 22 paid courses generated 100% of direct platform revenue (\$227,238.89). Furthermore, instructor workload analysis revealed that two instructors accounted for 60.86% of all platform transactions. These findings provide actionable decision support for proactive portfolio planning, pricing adjustments, marketing capital allocation, and instructor resource optimization.

Keywords: Predictive Analytics, Course Demand Forecasting, Revenue Forecasting, Machine Learning, Random Forest, Gradient Boosting, Educational Data Mining, Portfolio Management.

1 Introduction

The digital transformation of education has accelerated the expansion of e-learning platforms, giving millions of learners access to diverse academic and professional curricula. Commercial platforms like EduPro operate in a competitive market where learner preferences, industry skill demands, and monetization models evolve rapidly. To maintain financial stability and operational efficiency, e-learning providers must move beyond retrospective reporting and adopt proactive, data-driven planning systems.

Historically, online learning platforms have relied on descriptive dashboards to monitor key metrics, such as user acquisition, course completions, and gross revenue. While descriptive analytics provide visibility into past performance, they lack the forward-looking insights needed

for strategic planning. Without predictive models, platform administrators struggle to estimate future server bandwidth, allocate marketing budgets, evaluate course pricing structures, or balance instructor workloads. Predictive intelligence bridges this gap by using historical transaction data, user demographics, and instructor characteristics to project future demand and financial outcomes.

This study establishes a predictive analytics framework for the EduPro platform. By synthesizing multi-relational data across users, instructors, courses, and financial transactions, we develop course-level forecasting models for future enrollment counts and revenue generation. The resulting predictive infrastructure gives platform executives forward-looking guidance to optimize resource distribution, pricing strategies, and curriculum planning.

2 Problem Statement

EduPro offers a varied catalog of free and paid courses across several academic and technical subjects. Despite accumulating comprehensive transaction logs, the platform lacks integrated predictive models to guide operational decision-making. This absence of predictive analytics creates operational and strategic challenges across several key operational areas:

1. **Course Enrollment Demand Uncertainty:** Platform administrators lack reliable estimates of future course enrollments, leading to misallocated marketing spends and unbalanced server infrastructure.
2. **Revenue Planning Invisibility:** Financial forecasting is hindered by an inability to project course- and category-level revenues, creating budget uncertainty.
3. **Unassisted Course Launching:** New courses are often introduced based on instructor availability or subjective preference rather than quantitative demand signals.
4. **Static Pricing Strategies:** Course prices are assigned without an empirical understanding of how pricing attributes interact with historical financial performance.
5. **Instructor Workload Imbalances:** Platform transaction activity is heavily concentrated among a small subset of instructors, leaving key talent over-leveraged while underutilizing others.

Transforming raw historical transaction data into predictive intelligence is essential for EduPro to transition from passive historical reporting to proactive strategic planning.

3 Research Gap

While educational data mining literature extensively covers student dropout rates, learning analytics, and student performance modeling, predictive modeling focused on platform-level course economics remains limited. Existing e-learning business research often treats enrollment demand and financial revenue as tightly coupled, linear metrics. However, on freemium digital platforms, demand volume is structurally separated from direct revenue generation due to the presence of unpaid introductory courses.

Furthermore, prior e-learning economic studies often omit multi-entity platform interactions, such as instructor experience metrics, rating distributions, subject expertise alignment, and transaction concentration. There is also a lack of empirical research evaluating how ensemble machine learning techniques perform when simultaneously predicting student demand (which often exhibits narrow variance) versus financial revenue (which features high skewness and bimodal distribution). This paper addresses this research gap by developing a dual-target predictive modeling approach using validated platform transaction data.

4 Research Objectives

To address EduPro’s operational challenges and fill gaps in e-learning economic literature, this study establishes the following technical and analytical objectives:

1. **Data Architecture & Integrity Validation:** Validate relational schema and referential integrity across 3,000 users, 60 teachers, 60 courses, and 10,000 transaction logs.
2. **Feature Engineering Framework:** Formulate course-level metrics, including aggregate instructor ratings, subject expertise match rates, historical revenue performance, and price bands.
3. **Predictive Model Development & Optimization:** Train and benchmark machine learning algorithms (Linear Regression, Ridge, Lasso, Random Forest, Gradient Boosting) for two primary targets: Future Enrollment Count and Future Course Revenue.
4. **Model Performance & Evaluation Metrics:** Evaluate model performance using MAE, RMSE, R^2 , residual analysis, and percentage errors, accounting for statistical nuances like negative enrollment R^2 values.
5. **Predictive Feature Importance & Pricing Dynamics:** Identify top predictive features for revenue generation and analyze pricing correlations without asserting unverified causal claims.
6. **Freemium & Category Analysis:** Evaluate performance disparities between 38 free and 22 paid courses, as well as demand versus revenue trends across 12 academic categories.
7. **Strategic Portfolio Classification:** Classify all 60 platform courses into strategic decision quadrants to identify high-opportunity courses and candidates for strategic improvement.

5 Dataset and Data Preparation

The study utilizes data from the EduPro database. The relational schema comprises four primary entities: Users, Teachers, Courses, and Transactions. The user entity contains demographic information ($N = 3,000$ unique users, ages 15 to 35). The teacher entity tracks instructor profiles ($N = 60$ instructors, ages 27 to 50, experience ranging from 1 to 24 years, ratings between 1.05 and 4.97). The course entity details curriculum attributes ($N = 60$ courses across 12 categories, durations between 1.20 and 49.73 hours, ratings between 1.13 and 4.94). The transaction entity logs individual enrollments ($N = 10,000$ transaction events).

Data quality verification confirmed zero missing values across all entities, zero duplicate records, and complete referential integrity (0 invalid UserIDs, 0 invalid CourseIDs, 0 invalid TeacherIDs).

To enable predictive modeling, transaction records were partitioned temporally:

- **Historical Dataset:** Transactions prior to the forecast split period ($N = 7,551$ records), yielding \$684,084.58 in cumulative historical revenue.
- **Forecast/Test Dataset:** Transactions during the forecast observation period ($N = 2,449$ records), generating \$227,238.89 in actual future revenue.

Aggregating these transaction logs to the course level ($N = 60$ observations) established paired historical input features and future actual targets for each course.

Table 1: Dataset and Analytical Structure

Data Entity	Records	Key Attributes / Features	Data Type Range / Categories	Validation Result
Users	3,000	UserID, UserName, Age, Gender, Email	Age: 15–35 years; Categorical Gender	0 Invalid UserIDs
Teachers	60	TeacherID, TeacherName, Age, Expertise, Experience, Rating	Experience: 1–24 yrs; Rating: 1.05–4.97	0 Invalid TeacherIDs
Courses	60	CourseID, CourseName, Category, Type, Level, Price, Duration, Rating	Price: \$0.00–\$490.90; 12 Subject Categories	0 Invalid CourseIDs
Transactions	10,000	TransactionID, UserID, CourseID, Amount, PaymentMethod, TeacherID	Amount: \$0.00–\$490.90	0 Invalid Foreign Keys
Analytical Dataset	60	Merged Course-Level Historical & Feature-Engineered Variables	18 Input Features, 2 Target Variables	100% Integrity

6 Methodology

The predictive analytics workflow follows a systematic pipeline designed to transform raw operational data into actionable strategic insights.

6.1 EduPro Predictive Analytics Workflow Framework

The data preparation pipeline uses Scikit-Learn’s `ColumnTransformer` architecture. Categorical features (`CourseCategory`, `CourseType`, `CourseLevel`, `PriceBand`, `DurationBucket`, `RatingTier`, `ExperienceBucket`) are transformed using one-hot encoding. Continuous numerical features are standardized using z-score normalization via `StandardScaler`:

$$z = \frac{x - \mu}{\sigma} \quad (1)$$

Model accuracy is evaluated using three primary metrics:

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (2)$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (3)$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (4)$$

Where y_i represents the actual target value, $\hat{y}_i$ is the predicted model output, $\bar{y}$ is the mean of the actual target, and n is the number of evaluation observations ($n = 12$ for the 20% test split).

Table 2: EduPro Predictive Analytics Workflow Stages

Stage	Process Phase	Operational Focus & Technical Implementation
Stage 1	Data Ingestion	Extraction of relational tables: Users ($N = 3,000$), Teachers ($N = 60$), Courses ($N = 60$), and Transactions ($N = 10,000$).
Stage 2	Data Cleaning	Entity integrity validation, foreign key mapping verification, numerical range checks.
Stage 3	Feature Engineering	Aggregation of teacher metrics, expertise matching, temporal revenue and enrollment derivations.
Stage 4	Exploratory Data Analysis	Correlation analysis, price-type distributions, instructor transaction concentration checks.
Stage 5	Model Development	Training Linear Regression, Ridge, Lasso, Random Forest, and Gradient Boosting pipelines.
Stage 6	Model Evaluation	Benchmarking MAE, RMSE, and R^2 metrics; residual distribution analysis.
Stage 7	Demand & Revenue Forecasting	Generating course-level enrollment and revenue predictions across all 60 courses.
Stage 8	Business Decision Support	Course portfolio matrix classification, pricing review, instructor resource optimization.

7 Exploratory Data Analysis

Exploratory analysis of historical transaction records revealed several key structural characteristics of the platform.

7.1 Course Distribution and Pricing Categorization

The platform catalog consists of 60 courses divided into 38 Free courses (CoursePrice = \$0.00) and 22 Paid courses (CoursePrice ranging from \$0.78 to \$490.90, with a mean paid price of \$253.60). These courses span 12 academic categories, with exactly 5 courses per category.

7.2 Instructor Workload Concentration

Transaction analysis across instructors identified a high degree of workload concentration. Out of 60 instructors on the platform, two accounted for the majority of student transaction events:

- **TC00042 (Yolanda Levine):** Machine Learning instructor (21 years experience, rating 4.97), processed 3,061 transactions (30.61% of platform total).
- **TC00040 (Kimberly Miller):** Cybersecurity instructor (24 years experience, rating 4.58), processed 3,025 transactions (30.25% of platform total).

Together, these two instructors handled 6,086 transactions, representing 60.86% of all transaction volume. The remaining 58 instructors processed between 23 and 164 transactions each, with a mean of approximately 67.48 transactions. This imbalance reflects platform routing rules and default instructor assignments rather than lower teaching quality among the remaining staff.

8 Feature Engineering

To improve the predictive capacity of candidate models, 18 domain-specific features were constructed from the relational tables:

1. Teacher-Level Aggregations:

- **TeacherCount:** Count of unique instructors assigned to transactions for a given course.
- **AvgTeacherExperience:** Mean years of teaching experience across assigned instructors.
- **AvgTeacherRating:** Average rating of instructors leading the course.
- **TeacherExpertiseMatchRate:** Proportion of assigned teachers whose domain expertise matches the course's category.

2. Categorical Bins and Bounded Metrics:

- **PriceBand:** Segmented into Free (\$0), Low ($\leq \150), Medium ($\leq \$300$), and High ($> \300).
- **DurationBucket:** Segmented into Short (≤ 15 hrs), Medium (≤ 30 hrs), and Long (> 30 hrs).
- **RatingTier:** Segmented into Low (< 2.5), Medium (< 4.0), and High (≥ 4.0).
- **CourseLevelEncoded:** Integer mapping (Beginner = 1, Intermediate = 2, Advanced = 3).

3. Historical Performance Features:

- **PastEnrollmentCount:** Total student enrollments recorded prior to the forecast period.
- **PastRevenue:** Total revenue generated prior to the forecast period.
- **PastRevenuePerEnrollment:** Historical financial yield per enrolled student.

9 Predictive Modeling

Models were trained on 80% of the course-level observations ($N = 48$) and evaluated on a 20% test split ($N = 12$). Machine learning algorithms were developed separately for the two primary targets: Future Enrollment Count and Future Course Revenue.

Table 3: Final Model Performance Comparison

Target Variable	Candidate Model	Eval MAE	Eval RMSE	Eval R^2	Selection Status
Future Enrollment	Random Forest	6.130417	7.322472	-0.236758	Selected Best
Future Enrollment	Gradient Boosting	6.050173	7.697334	-0.366626	Benchmark Comparison
Future Enrollment	Lasso Regression	6.622926	7.920794	-0.447127	Benchmark Comparison
Future Enrollment	Ridge Regression	6.957012	8.078633	-0.505376	Benchmark Comparison
Future Enrollment	Linear Regression	6.680913	8.624361	-0.715628	Benchmark Comparison
Future Revenue	Gradient Boosting	814.003007	1457.664702	0.960063	Selected Best
Future Revenue	Random Forest	875.619483	1473.665661	0.959182	Benchmark Comparison
Future Revenue	Ridge Regression	971.391792	1508.623229	0.957222	Benchmark Comparison
Future Revenue	Lasso Regression	999.765639	1576.576024	0.953282	Benchmark Comparison
Future Revenue	Linear Regression	1011.731212	1595.792205	0.952136	Benchmark Comparison

9.1 Enrollment Model Interpretation

Random Forest yielded the lowest RMSE (7.322472) and the highest R^2 (-0.236758) among the evaluated enrollment models, while Gradient Boosting achieved the lowest MAE (6.050173). However, the Random Forest model's R^2 score was negative ($R^2 = -0.236758$), indicating weak course-level out-of-sample explanatory power. In regression analysis, a negative R^2 occurs when a model performs worse than a simple horizontal line corresponding to the target mean.

This negative R^2 stems from low variance in the underlying course-level enrollment target (actual future enrollments range narrowly between 22 and 54 students, with a mean of 40.816667 and a standard deviation of 6.654912). When the total variance of the actual target is small, even small absolute prediction errors produce a negative R^2 value. Despite the negative R^2 on the held-out test set, the full 60-course aggregate prediction analysis produced an average absolute error of approximately 2.90 students per course, supporting its use for platform-level planning.

9.2 Revenue Model Interpretation

The Gradient Boosting model performed well for course revenue forecasting, achieving an R^2 score of 0.960063, an MAE of \$814.003007, and an RMSE of \$1,457.664702. The high R^2 reflects the strong separation between \$0 revenue free courses and high-revenue paid courses, alongside strong predictive signals from historical financial features.

10 Model Evaluation and Feature Importance

Analyzing feature importances in the optimal Gradient Boosting revenue model highlights the primary factors driving financial forecasts on the platform.

Table 4: Feature Importance Analysis (Gradient Boosting Revenue Model)

Rank	Feature Name	Feature Description	Relative Importance	Cumulative Weight
1	PastRevenue	Cumulative historical course revenue prior to forecast period	49.97%	49.97%
2	CoursePrice	Listed nominal price for course enrollment	25.35%	75.32%
3	PastRevenuePerEnrollment	Historical financial yield per enrolled student	20.44%	95.76%
4	AvgTeacherRating	Mean student rating across assigned instructors	2.01%	97.77%
5	AvgTeacherExperience	Mean teaching experience (years) of instructors	1.66%	99.43%
6	CourseRating	Public student rating of course curriculum	0.35%	99.78%
7	All Other Features (12)	Remaining engineered and categorical features	0.22%	100.00%

The top three historical financial variables (PastRevenue, CoursePrice, PastRevenuePerEnrollment) account for 95.76% of the revenue model's predictive weight.

10.1 Interpretation of Feature Importance and Pricing Dynamics

Feature importance metrics reflect predictive association within the decision tree structure rather than causal relationships. While `CoursePrice` shows a strong positive correlation with predicted revenue (≈ 0.9902), this association arises because paid courses generate direct cash inflows whereas free courses generate \$0 direct revenue. This statistical association does not imply that unilaterally increasing course prices will automatically increase total revenue.

11 Results and Forecast Analysis

Applying the selected models across the platform generated course-level predictions for all 60 offerings.

11.1 Platform-Level Aggregate Forecast Comparison

At the platform level, aggregated model predictions aligned closely with actual future outcomes.

Table 5: Actual vs. Predicted Aggregate Platform Forecasts

Tar- get	Actual Total	Predicted Total	Abs. Variance	% Diff	Bias Direction	Aggregate Forecast Accu
Fu- ture En- roll- ment	2,449.00 stds	2,464.385 stds	15.385 stds	0.6282%	+15.385 (Over)	99.37
Fu- ture Rev- enue	\$227,238.89	\$222,744.35	\$4,494.54	1.9779%	-\$4,494.54 (Under)	98.02

At the aggregate level, enrollment forecasts achieved 99.371784% aggregate forecast accuracy (+15.385 students bias), while revenue forecasts achieved 98.022109% aggregate forecast accuracy (-\$4,494.54 bias).

This negative R^2 stems from low variance in the underlying course-level enrollment target (actual future enrollments range narrowly between 22 and 54 students, with a mean of 40.816667 and a standard deviation of 6.654912). When the total variance of the actual target is small, even small absolute prediction errors produce a negative R^2 value. Despite the negative R^2 on the held-out test set, the full 60-course aggregate prediction analysis produced an average absolute error of approximately 2.90 students per course, supporting its use for platform-level planning.

11.2 Revenue Model Interpretation

The Gradient Boosting model performed well for course revenue forecasting, achieving an R^2 score of 0.960063, an MAE of \$814.003007, and an RMSE of \$1,457.664702. The high R^2 reflects the strong separation between \$0 revenue free courses and high-revenue paid courses, alongside strong predictive signals from historical financial features.

12 Model Evaluation and Feature Importance

Analyzing feature importances in the optimal Gradient Boosting revenue model highlights the primary factors driving financial forecasts on the platform.

Table 6: Feature Importance Analysis (Gradient Boosting Revenue Model)

Rank	Feature Name	Feature Description	Relative Importance	Cumulative Weight
1	PastRevenue	Cumulative historical course revenue prior to forecast period	49.97%	49.97%
2	CoursePrice	Listed nominal price for course enrollment	25.35%	75.32%
3	PastRevenuePerEnrollment	Historical financial yield per enrolled student	20.44%	95.76%
4	AvgTeacherRating	Mean student rating across assigned instructors	2.01%	97.77%
5	AvgTeacherExperience	Mean teaching experience (years) of instructors	1.66%	99.43%
6	CourseRating	Public student rating of course curriculum	0.35%	99.78%
7	All Other Features (12)	Remaining engineered and categorical features	0.22%	100.00%

The top three historical financial variables (PastRevenue, CoursePrice, PastRevenuePerEnrollment) account for 95.76% of the revenue model's predictive weight.

12.1 Interpretation of Feature Importance and Pricing Findings

Feature importance metrics reflect predictive association within the decision tree structure rather than causal relationships. While CoursePrice shows a strong positive correlation with predicted revenue (≈ 0.9902), this association arises because paid courses generate direct cash inflows whereas free courses generate \$0 direct revenue. This statistical association does not imply that unilaterally increasing course prices will automatically increase total revenue.

13 Results and Forecast Analysis

Applying the selected models across the platform generated course-level predictions for all 60 offerings.

13.1 Platform-Level Aggregate Forecast Comparison

At the platform level, aggregated model predictions aligned closely with actual future outcomes.

At the aggregate level, enrollment forecasts achieved 99.371784% accuracy (+15.385 students bias), while revenue forecasts achieved 98.022109% accuracy (-\$4,494.54 bias).

Table 7: Actual vs. Predicted Aggregate Platform Forecasts

Target	Actual Total	Predicted Total	Abs. Variance	% Diff	Bias Direction	Accuracy
Future Enrollment	2,449.00 stds	2,464.385 stds	15.385 stds	0.6282%	+15.385 (Over)	99.3718%
Future Revenue	\$227,238.89	\$222,744.35	\$4,494.54	1.9779%	-\$4,494.54 (Under)	98.0221%

13.2 Course-Level Errors and Residual Distributions

Evaluating prediction errors at the course level provides a clearer picture of model performance:

- **Enrollment Residuals:** Average Enrollment Absolute Error = 2.903917 students; Average Enrollment Percentage Error = 7.739974%. Residuals ranged from -13.285 to +9.605 students (mean = -0.256417 , std = 4.069111, median = -0.125). Across the 60 courses, 29 were under-predicted and 31 were over-predicted.
- **Revenue Residuals:** Average Revenue Absolute Error = \$162.800882; Average Revenue Percentage Error = 1.844030%. On the test set, revenue residuals ranged from -\$1,469.47 to +\$4,470.11 (mean = \$374.563840, std = \$1,471.358672, median = \$0.003567), with 6 courses under-predicted and 6 courses over-predicted.

13.3 Freemium Dynamics: Free vs. Paid Course Comparison

Segmenting performance by pricing structure highlights freemium patterns on the platform.

Table 8: Free vs. Paid Course Performance Matrix

Metric Parameter	Free Segment (N=38)	Paid Segment (N=22)	Platform Total (N=60)
Actual Future Enrollment	1,571 students	878 students	2,449 students
Predicted Future Enrollment	1,564.755 students	899.630 students	2,464.385 students
Enrollment Variance	-6.245 students	+21.630 students	+15.385 students
Actual Future Revenue	\$0.00	\$227,238.89	\$227,238.89
Predicted Future Revenue	\$0.078393	\$222,744.27	\$222,744.35
Revenue Variance	+\$0.078393	-\$4,494.62	-\$4,494.54

Free courses drive 64.15% of total future enrollments (1,571 students) but generate \$0.00 in direct revenue. Conversely, the 22 paid courses account for 100% of platform revenue (\$227,238.89) from 35.85% of total enrollments (878 students).

13.4 Category-Level Forecast Analysis

Artificial Intelligence, Business, and Project Management generated the highest revenue totals (\$52,438.14, \$46,278.51, and \$44,356.45 actual, respectively). Categories such as Data Science,

Table 9: Category-Level Actual vs. Predicted Results

Course Category	Courses	Actual Enr.	Pred. Enr.	Actual Rev.	Category Profile
Artificial Intelligence	5	208	204.315	\$52,438.14	High Yield / High Revenue
Business	5	209	208.490	\$46,278.51	High Yield / High Revenue
Project Management	5	208	221.080	\$44,356.45	High Demand & High Revenue
Data Science	5	218	218.075	\$17,577.97	High Demand / Moderate Yield
Cybersecurity	5	201	200.285	\$16,164.66	Moderate Revenue Yield
Programming	5	197	205.455	\$17,946.64	Moderate Revenue Yield
Design	5	216	217.890	\$10,778.43	High Demand / Lower Yield
Digital Marketing	5	186	186.210	\$10,910.35	Moderate Demand & Yield
Finance	5	210	207.965	\$6,536.24	High Demand / Lower Yield
Web Development	5	206	199.745	\$4,228.10	Lower Monetization Yield
Machine Learning	5	183	185.330	\$23.40	Loss-Leader / Introductory
Marketing	5	207	209.545	\$0.00	Pure Acquisition Engine

Design, and Finance recorded strong student enrollments (218, 216, and 210 students) but lower direct revenue due to a higher proportion of free course offerings.

13.5 Course Opportunity Rankings

Table 10: Top 10 Revenue Opportunities

Rank	Course ID	Course Name	Category	Type	Price	Pred. Rev.
1	CR00050	Computer Vision	Artificial Intelligence	Paid	\$490.90	\$21,108.70
2	CR00012	Business Strategy	Business	Paid	\$487.92	\$20,980.56
3	CR00034	Data Encryption	Cybersecurity	Paid	\$394.26	\$16,712.85
4	CR00001	Python Basics	Programming	Paid	\$472.28	\$16,496.11
5	CR00040	Leadership in Projects	Project Management	Paid	\$327.99	\$15,743.50
6	CR00049	Deep Learning	Artificial Intelligence	Paid	\$415.51	\$15,474.37
7	CR00039	Project Planning	Project Management	Paid	\$346.71	\$14,908.54
8	CR00038	Risk Management	Project Management	Paid	\$342.61	\$13,704.41
9	CR00011	Entrepreneurship 101	Business	Paid	\$326.02	\$13,040.79
10	CR00025	Statistical Analysis	Data Science	Paid	\$299.57	\$12,282.37

Course CR00040 (Leadership in Projects) stands out by ranking in the top 10 for both enrollment demand (46,235 students) and revenue generation (\$15,743.50).

14 Research Interpretation

Synthesizing empirical findings yields key insights into platform economics and model evaluation.

Table 11: Top 10 Enrollment Opportunities

Rank	Course ID	Course Name	Category	Type	Price	Pred. Enr.
1	CR00043	Corporate Finance	Finance	Free	\$0.00	48.610
2	CR00010	3D Modeling	Design	Free	\$0.00	47.475
3	CR00003	C++ for Beginners	Programming	Free	\$0.00	47.195
4	CR00036	Agile Project Management	Project Management	Free	\$0.00	46.900
5	CR00040	Leadership in Projects	Project Management	Paid	\$327.99	46.235
6	CR00021	Data Analysis with Python	Data Science	Free	\$0.00	46.150
7	CR00007	UI/UX Design	Design	Free	\$0.00	46.005
8	CR00058	Social Media Strategy	Digital Marketing	Free	\$0.00	44.835
9	CR00030	Reinforcement Learning	Machine Learning	Free	\$0.00	44.570
10	CR00055	Full-Stack Web Dev	Web Development	Paid	\$57.32	44.475

Table 12: Core Research Findings and Strategic Implications

#	Empirical Finding	Analytical Interpretation	Strategic Implication
1	Revenue Model Superiority ($R^2 = 0.9601$)	High explained variance due to bimodal pricing split and strong financial signals	High confidence in forward revenue projections
2	Historical Financial Primacy (95.76% top 3 weight)	Historical revenue and course price dominate tree-based predictive weight	Past course yield strongly predicts future financial performance
3	Non-Causal Price Association (≈ 0.9902 corr)	Reflects pricing tier segment split, not price elasticity	Price increases must be tested cautiously rather than assumed beneficial
4	Decoupled Demand vs. Revenue Signal	Course popularity does not directly mirror monetization yield	Dual-target forecasting is required for freemium platforms
5	Freemium Traffic Engine (64.15% enrollment share)	Free courses drive enrollment volume but produce \$0 direct revenue	Utilize free courses as lead generators for paid conversion
6	Monetization Concentration (22 paid courses = 100% rev)	Revenue generation is concentrated in a limited paid catalog	Focus marketing capital on high-yield paid offerings
7	Low Enrollment Variance Nuance ($R^2 = -0.2368$)	Constrained enrollment range causes negative R^2 despite low MAE (6.13)	Evaluate multiple metrics alongside R^2 for low-variance targets
8	Instructor Concentrated Routing (60.86% share)	Two instructors handle most transactions due to system assignment rules	Diversify instructor assignments to mitigate operational risk
9	Subject Domain Asymmetry	AI/Business drive revenue; Design/Finance drive traffic volume	Align marketing strategies with category strengths
10	Portfolio Optimization Utility	Predictive models successfully segment 60 courses into decision quadrants	Enables data-driven curriculum strategy

14.1 Business Insights

To translate research outputs into operational strategy, EduPro can implement a decision-support framework that categorizes courses into portfolio quadrants based on predicted performance.

14.2 Decision-Support Framework Pipeline

Table 13: EduPro Decision-Support Framework

Historical Ingestion	Predictive ML	Portfolio Classification	Execution Plan
• Users ($N = 3,000$) • Teachers ($N = 60$) • Courses ($N = 60$) • Transactions ($N = 10,000$)	• Random Forest Demand Model • Gradient Boosting Revenue Model • Feature Importance Analysis	• High Demand / High Rev (15) • High Demand / Low Rev (15) • Low Demand / High Rev (15) • Low Demand / Low Rev (15)	• Scale Marketing Capital • Test Monetization Pathways • Rebalance Instructors • Restructure Low-Performing Priorities

14.3 Research-to-Business Insight Synthesis

Table 14: Research-to-Business Insight Synthesis

Research Phase	Analytical Finding	Business Insight	Operational Action
Data Audit	Two instructors process 60.86% of platform transactions	Operational bottleneck and key-person dependency risk	Rebalance transaction assignment rules across qualified instructors
Model Eval	Revenue $R^2 = 0.9601$; Enrollment MAE = 6.130417	High financial predictability alongside steady enrollment patterns	Deploy forward-looking cash flow and server capacity models
Feature Drivers	Past revenue and price account for 75.32% of revenue model weight	Financial history and pricing tiers dominate revenue forecasts	Conduct controlled pricing experiments across paid courses
Freemium Analysis	Free courses account for 64.15% of enrollments	Free courses attract users but do not generate direct revenue	Design conversion funnels linking free offerings to paid certificates
Portfolio Audit	15 High-Demand/High-Revenue courses vs. 15 Low-Demand/Low-Revenue courses	A large portion of the catalog requires strategic intervention, alongside the 3-course High-Opportunity set	Protect core revenue drivers while auditing low-performing offerings

15 Strategic Recommendations

Based on empirical results, four strategic initiatives are recommended for EduPro:

1. **Scale High-Opportunity Offerings:** Direct marketing resources toward core revenue drivers: CR00040 (Leadership in Projects), CR00006 (Graphic Design Basics), and CR00024 (Big Data Analytics). These 3 high-opportunity offerings combine high predicted demand with strong financial yield.
2. **Monetize High-Enrollment Free Offerings:** High-volume free courses—such as CR00043 (Corporate Finance, 48.610 predicted enrollments) and CR00010 (3D Modeling, 47.475 predicted enrollments)—should serve as conversion funnels. Introducing paid completion certificates or advanced sequel modules can convert user volume into direct revenue.
3. **Address Improvement-Priority Courses:** Audit low-performing offerings (such as Java Programming, C++ for Beginners, Advanced Python, Photoshop Mastery). Updating curriculum content, refreshing promotional material, or bundling courses can help improve enrollment and engagement.
4. **Rebalance Instructor Workload Distribution:** Redesign transaction assignment algorithms to reduce operational reliance on instructors TC00042 and TC00040. Distributing workload across the platform's 58 other qualified instructors will mitigate operational risk and improve platform scalability.

16 Limitations

Several limitations should be considered when interpreting these findings:

- **Sample Size Constraints:** The modeling dataset consists of 60 aggregated course observations derived from 10,000 transaction records.
- **Narrow Enrollment Target Variance:** Course-level enrollments exhibit low variance (range 22–54 students), contributing to negative R^2 values despite low absolute prediction errors.
- **Unobserved External Factors:** External marketing spend, ad campaign details, and macro-economic factors were unavailable in the dataset.
- **Platform-Specific Context:** Results reflect the operational structure of EduPro and may not apply directly to other e-learning ecosystems.
- **Non-Causal Feature Importances:** Feature importances and correlation metrics (≈ 0.9902 for course price) represent predictive associations within the model, not causal price elasticity.

17 Conclusion

This study demonstrates the application of predictive machine learning models for course demand and revenue forecasting on the EduPro platform. By evaluating relational data across 3,000 users, 60 instructors, 60 courses, and 10,000 transactions, the research developed accurate course-level forecasting models. Random Forest yielded the lowest RMSE and highest R^2 among evaluated enrollment models (RMSE = 7.322472, MAE = 6.130417 students, platform aggregate forecast agreement = 99.371784%), while Gradient Boosting achieved the lowest MAE (MAE = 6.050173 students) for enrollment demand and demonstrated high precision for course revenue forecasting (R^2 = 0.960063, MAE = \$814.003007, platform aggregate forecast agreement = 98.022109%).

The empirical findings highlight a fundamental freemium platform dynamic: enrollment volume is decoupled from direct revenue generation. While 38 free courses drive 64.15% of platform enrollment volume, 22 paid courses generate 100% of direct revenue. Furthermore,

historical financial metrics account for 95.76% of predictive weight in revenue modeling. By combining predictive demand and revenue forecasts, e-learning platform executives can transition from descriptive monitoring to proactive portfolio management, optimizing marketing spend, pricing tiers, and instructor allocations.